

Harnessing cassava-based ethanol for rural communities: A conditional circular pathway to reduce biomass dependence in rural Congo[☆]

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ABSTRACT

Cooking energy in rural areas of Kinshasa Province, Democratic Republic of Congo, relies almost entirely on traditional biomass, generating persistent health burdens, forest degradation, and physical drudgery in the absence of modern cooking alternatives at the village level. Although numerous clean cooking initiatives have been implemented, limited attention has been paid to informal, locally embedded biomass transformation systems. This study addresses this gap by examining the sustainability potential of cassava-based ethanol in Bita, a rural community on the Bateke Plateau. Drawing on empirical fieldwork conducted among 287 rural households, the study combines a cradle-to-gate Life Cycle Assessment framework (LCA) and a Multi-Criteria Decision Analysis (MCDA) to compare two ethanol production pathways—an artisanal cassava–maize system and a field-replicated cassava-only pathway—and to assess ethanol alongside firewood and charcoal.

The analysis evaluates whether locally produced ethanol can reduce key burdens associated with biomass dependence under realistic rural conditions. Results show that ethanol achieves a higher composite MCDA score (3.19) than firewood (2.62) and charcoal (2.71), reflecting improvements in health-related performance, reduced physical effort, and lower pressure on forest resources. However, these gains are accompanied by trade-offs related to water intensity, production efficiency, and scalability, particularly under seasonal water constraints and limited institutional support. Overall, cassava-based ethanol emerges as a context-dependent niche pathway within biomass-dependent energy systems rather than a universal solution for rural energy transitions.

Introduction

Biomass remains the primary energy source for communities in developing countries, supplying fuel for essential domestic and productive activities. In sub-Saharan Africa, traditional fuels—mainly firewood and charcoal—continue to dominate household energy use due to limited access to modern energy services (IEA, 2022; WHO, 2025). While culturally embedded, their conventional use entails substantial external costs, including indoor air pollution, health risks, forest degradation, and contributions to climate change (Bonjour et al., 2013; WHO, 2025). Past interventions have largely focused on incremental technological gains, such as improved cookstoves. However, by addressing combustion efficiency without challenging underlying biomass dependence, such interventions have failed to deliver durable improvements in health, livelihoods, or environmental outcomes (Puzzolo et al., 2016). Increasingly, the literature emphasizes that

sustainable biomass transitions require a systemic perspective that considers feedstock sourcing, production practices, and livelihood linkages rather than combustion performance alone (Creutzig et al., 2015; Köhlin et al., 2011). This need is particularly evident in contexts where biomass use is structurally embedded in informal systems.

The Democratic Republic of Congo (DRC) exemplifies this situation. Over 95% of households rely on firewood and charcoal for cooking, contributing to deforestation and rural economic vulnerability (World Bank, 2023). At the same time, the DRC hosts a long-standing but underexplored system of artisanal biomass transformation: locally produced ethanol, commonly known as *lotoko*. Distilled primarily from cassava and maize using informal methods, *lotoko* is produced exclusively for beverage consumption and is socially framed as an alcoholic drink rather than as an energy carrier. As a result, its potential use as a cooking fuel remains largely unknown to policymakers and is absent from national energy statistics and rural clean cooking strategies (CCAC,

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2024). This institutional invisibility reflects not a lack of socio-economic relevance, but a lack of empirical assessment of ethanol as an energy option, rendering a structurally important biomass-based system invisible to energy policy.

Recent work in Bita village (Tshimungu et al., 2025) demonstrated that biomass dependence in rural Kinshasa is structurally reinforced by poverty, spatial constraints, and seasonal scarcity, despite strong household aspirations for transition. That study identified cassava-based ethanol as a socio-technical niche innovation with promising combustion performance, high social acceptability, and near-cost willingness to pay. However, while the socio-technical feasibility of ethanol was established, its broader environmental performance and systemic sustainability remained unexamined. The present study builds directly on these findings by assessing whether cassava-based ethanol can reduce the environmental and structural burdens associated with biomass reliance under realistic rural production conditions.

Adopting a socio-technical systems perspective, this study conceptualizes cooking fuels as configurations of resources, practices, actors, and institutions embedded in specific local contexts (Geels, 2004; Sovacool, 2016). Sustainability outcomes therefore depend not only on technical efficiency, but on how production and use are integrated into livelihoods and material cycles. Consequently, circularity is approached as an embedded system property—reflected, for instance, in the local reuse of distillation residues as soil amendments or construction inputs—rather than as a fully quantifiable metric (HLPE, 2020; Kirchherr et al., 2017). In this study, the notion of “conditional circularity” refers to informal residue reuse practices that generate localized material recirculation benefits under specific rural conditions, while remaining constrained by technical, environmental, and institutional limitations.

Within this framework, cassava-based ethanol represents a distinctive but context-dependent biomass valorization pathway. Cassava is non-seasonal, resilient to climatic variability, and subject to high post-harvest losses, ensuring a relatively stable feedstock compared to maize, which faces recurrent food security constraints (FAO, 2017; Vutula, 2024). While shifting toward a cassava-only pathway could enhance local value addition, artisanal ethanol production itself relies on traditional biomass for distillation, raising critical questions about net environmental performance and local environmental risks. In addition, distillation generates liquid effluents (vinasse or stillage) that, if discharged untreated, may contribute to localized water deoxygenation or soil acidification under rural conditions.

This study addresses these gaps by combining Life Cycle Assessment (LCA) and Multi-Criteria Decision Analysis (MCDA). The LCA explicitly accounts for energy and material inputs required for ethanol production, including biomass used for distillation, in order to assess net impacts across the production cycle. The overall objective is to determine whether a locally produced, cassava-based ethanol system can contribute to more sustainable biomass use across multiple dimensions. Specifically, the study addresses three research questions: (1) What are the main environmental impact drivers of artisanal and cassava-only ethanol production pathways in rural DRC? (2) How does cassava-based ethanol compare with firewood and charcoal across environmental, technical, economic, and socio-contextual dimensions?

(3) To what extent can decentralized ethanol production contribute to circular biomass use, and what limitations—such as effluent-related environmental risks—remain unresolved?

Methodology

Study design and methodological approach

This study employs a mixed-method approach to assess the sustainability of cassava-based ethanol as a household cooking fuel under rural conditions in the Democratic Republic of Congo (DRC). Life Cycle Assessment (LCA) is used to quantify and compare the environmental

impacts of alternative ethanol production pathways, while Multi-Criteria Decision Analysis (MCDA) is applied to compare cassava-based ethanol with dominant cooking fuels—firewood and charcoal—across multiple sustainability dimensions. The methods are applied complementarily: while the LCA focuses on the environmental trade-offs associated with feedstock substitution and artisanal transformation processes at the production stage, the MCDA encompasses the full energy service—including household cooking—to evaluate the overall sustainability of cassava-based ethanol against traditional woodfuels under rural conditions.

Study area and data collection context

The study was conducted in Bita, a village located on the outskirts of Kinshasa Province, Democratic Republic of Congo (DRC). Although administratively part of the capital region, Bita displays characteristics typical of rural areas in the country, including heavy reliance on firewood and charcoal for cooking, limited access to modern energy services, and predominantly informal, small-scale economic activities. Artisanal distillation of locally produced ethanol (lotoko) is practiced as a livelihood activity embedded in local resource cycles.

Primary data were collected in Bita (Fig. 1.) from operational artisanal ethanol production site to document real-world production conditions through direct observation. Batch-level measurements were conducted during substrate preparation, fermentation, and distillation to record feedstock quantities, water use, biomass consumption for distillation, ethanol yields, and residue management practices. In addition to documenting existing practices, an experimental ethanol production was conducted at the same sites using the same locally available equipment but applying a modified production procedure. This experimental replication enabled a direct comparison with observed artisanal production under identical environmental and technical conditions. Informal discussions with producers were used to clarify routine practices, batch variability, and local reuse of by-products. Together, observational and experimental data formed the empirical basis for modeling the two ethanol production pathways in the LCA and for informing socio-contextual, systemic, and circularity-related criteria in the MCDA.

Ethanol production pathways assessed

Two ethanol production pathways representative of existing local practice and a potential alternative were assessed. Their characterization builds on empirical field data previously collected and analyzed in Tshimungu et al. (2025) and is here extended to support a life cycle and multi-criteria sustainability assessment. Whereas the previous study focused on household behavior, structural lock-in, and socio-technical feasibility, this article evaluates production-stage environmental impacts and integrates multi-dimensional sustainability criteria to determine whether ethanol represents a structurally viable transition pathway. The first pathway corresponds to the artisanal cassava–maize ethanol system, which reflects the dominant production method in the study area. Cassava roots and maize grains are manually processed, mixed with water, and fermented using naturally occurring yeasts under ambient conditions. Distillation is performed using improvised metal stills heated with firewood, with limited process control, resulting in variable ethanol yields and purity. Solid residues and peels are typically reused locally, while liquid effluents are partially reused or discharged. The second pathway represents an alternative ethanol production system replicated under field conditions using locally available equipment but applying a modified production procedure. Fermentation follows a standardized duration with controlled inoculation, while distillation remains wood-fired and subject to temperature variability. This pathway was designed to assess whether modifying production practices—while maintaining local technical constraints—could improve system sustainability without compromising feasibility under rural

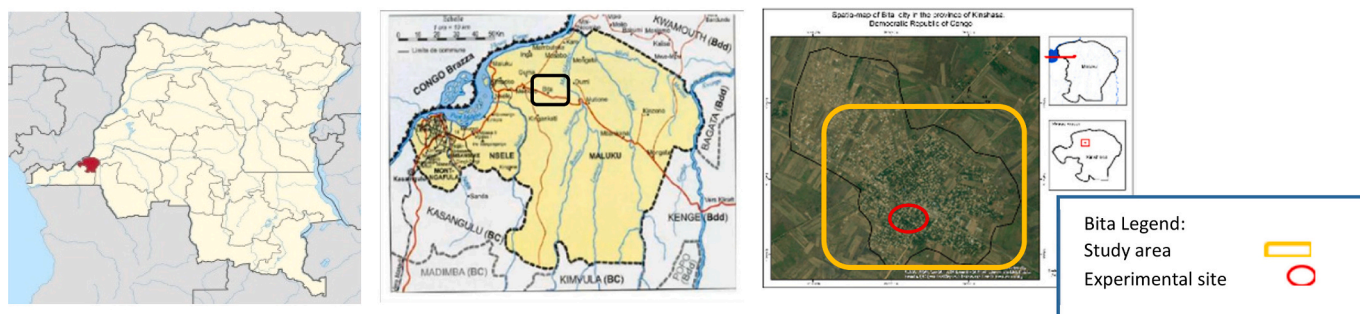


Fig. 1. Bitá Map (Source: Author's elaboration based on data from the Ministry of Environment, 2021).

conditions.

Life Cycle Assessment (LCA)

Goal and scope

The goal of the LCA is to identify and compare the dominant environmental impact drivers of the two ethanol production pathways under realistic rural production conditions. The functional unit is defined as 1 l of ethanol produced. The system boundary follows a cradle-to-gate approach, encompassing feedstock production, substrate preparation, fermentation, and distillation. Distribution and end-use combustion are excluded, as the LCA is designed to isolate production-stage impacts and enable a controlled comparison between alternative ethanol production pathways. This boundary choice has two important implications. First, the results should not be interpreted as representing the full life-cycle performance of ethanol as a cooking fuel, but rather as comparative production-stage indicators. Second, differences in combustion performance, thermal efficiency, and household-level emissions are intentionally addressed in the subsequent MCDA in order to avoid double counting and ensure analytical consistency across assessment stages. The LCA therefore provides a conservative and internally consistent estimate of environmental burdens associated with ethanol production under rural conditions. No allocation was applied to co-products; all residues were treated as waste to avoid over-crediting circularity benefits that are informal, context-specific, and difficult to quantify. Land-use change, transport, and capital goods were excluded due to high uncertainty and limited empirical relevance in the local context.

Palm kernel shells (PKS) were used opportunistically in both the artisanal and replication pathways and did not displace firewood, as the producer consistently purchased 10 kg of firewood per production cycle regardless of PKS use. As PKS were non-substitutive and present in both systems, their exclusion does not bias the comparative cradle-to-gate life cycle assessment. Broader sustainability dimensions are subsequently addressed through the integrated MCDA framework.

Due to the absence of DRC-specific life cycle inventory data for cassava cultivation and ethanol production, Brazilian proxy datasets were used to represent agricultural inputs and emission profiles. While these datasets provide structured and internally consistent inventories, they may not fully reflect local agronomic conditions, particularly with respect to pesticide intensity, fertilizer application rates, and yield variability. Results should therefore be interpreted as structurally comparative rather than as precise quantifications of site-specific environmental burdens. This is particularly relevant for toxicity-related categories, which are strongly influenced by agricultural proxy datasets. Residues were conservatively treated as waste without allocation of environmental credits. This assumption avoids overestimating circularity benefits but may underestimate potential environmental gains under optimized residue management scenarios. Land-use change, transport, and capital goods were excluded due to high uncertainty and the small-scale, locally embedded character of production. While this likely yields conservative midpoint estimates for certain impact

categories, it may underestimate impacts under large-scale expansion scenarios.

Life cycle inventory data and assumptions

Life cycle inventory (LCI) data were primarily derived from batch-level field measurements collected in Bitá and normalized to the functional unit of 1 l of ethanol produced. Primary data included feedstock quantities, water use, firewood consumption for distillation, ethanol yield, residue generation, and process-related emissions.

Due to the absence of country-specific datasets for the Democratic Republic of Congo, agricultural upstream processes were modeled using proxy datasets from established databases. Maize production was represented by *Maize, grain, global (GLO)* from Ecoinvent, while cassava cultivation was modeled using *Cassava, at farm, Brazil (BR)* from the Agri-footprint database. These proxies were applied consistently across pathways to preserve structural comparability.

Minor inputs lacking robust datasets, including palm kernel shells and malt, were excluded due to their marginal contribution and non-substitutive role in the production process. Emissions from wood-fired distillation were assumed identical across pathways, reflecting the use of comparable equipment under similar operating conditions. Biomass-related CO₂ emissions were treated as biogenic in accordance with standard LCA practice.

Tables 1 and 2 summarize the LCI inputs and outputs per functional unit for the artisanal and replication pathways, respectively.

Impact assessment method

The life cycle impact assessment (LCIA) was conducted in accordance with ISO 14040 and ISO 14044 standards using SimaPro 9.5. Environmental impacts were calculated using the ReCiPe 2016 Midpoint (H) method. The analysis focuses on midpoint impact categories relevant to rural biomass energy systems: Climate Change (CC), Particulate Matter Formation (PMF), Human Toxicity (HT), Terrestrial and Freshwater Ecotoxicity, Water Use (WU), Land Use (LU), and Biomass

Table 1

Life cycle inventory input-outputs per functional unit of ethanol produced (artisanal ethanol).

Parameters	Quantity	Unit
Cassava	1.24	kg/l
Maize	1.17	kg/l
Water	8	l/l
Firewood	0.48	kg/l
Fermentation residues	1.82	kg/l
Wastewater	8	l/l
Carbon dioxide (CO ₂), Biogenic	0.787	kg/l
Methane (CH ₄), Biogenic	1.78	g/l
Nitrous oxide (N ₂ O)	0.053	g/l
Carbon monoxide (CO)	38.8	g/l
Particulate matter (PM _{2.5})	3.5	g/l
Oxides of nitrogen (NO _x)	0.77	g/l
Volatile organic carbon (VOC)	4.8	g/l

Table 2

Life cycle inventory input-outputs per functional unit of ethanol produced (ethanol, replication).

Parameters	Quantity	Unit
Cassava	3.33	kg/l
Maize		
Yeast	33.3	g/l
Water	11	l/l
Firewood	0.48	kg/l
Fermentation residues	2.5	kg/l
Wastewater	11.12	l/l
Carbon dioxide (CO ₂), Biogenic	0.787	kg/l
Methane (CH ₄), Biogenic	1.78	g/l
Nitrous oxide (N ₂ O)	0.053	g/l
Carbon monoxide (CO)	38.8	g/l
Particulate matter (PM _{2.5})	3.5	g/l
Oxides of nitrogen (NO _x)	0.77	g/l
Volatile organic carbon (VOC)	4.8	g/l

Resource Depletion. Midpoint indicators were prioritized to reduce uncertainty and facilitate transparent comparison between production pathways. Endpoint results were used selectively to support interpretation of aggregated damage trends.

Multi-Criteria Decision Analysis (MCDA)

The MCDA constitutes an ex ante appraisal of household cooking fuel options under current rural conditions in Kinshasa. The analysis is designed to support context-specific decision-making by comparing fuels that are realistically available to households rather than idealized or future scenarios. Three options were assessed: firewood, charcoal, and locally produced ethanol. Ethanol was represented by the production pathway identified as environmentally preferable based on the LCA results, in order to ensure consistency between environmental assessment and decision analysis. All options reflect decentralized production and use systems typical of the study area, Bita, located on the Bateke Plateau.

Criteria selection and weighting

Evaluation criteria were structured into six sustainability dimensions: environmental, technical, economic, health, socio-contextual, and systemic/circularity-related. This framework was designed to ensure conceptual exhaustiveness while minimizing redundancy across indicators. Environmental criteria capture quantifiable biophysical impacts informed by LCA results for ethanol and complemented by literature-based emission factors and comparative benchmarks for firewood and charcoal. Health criteria address household exposure and safety risks associated with fuel use that are not fully represented by production-stage life cycle indicators. Systemic and circularity-related criteria reflect structural properties of local energy systems—such as forest dependence, reliance on external inputs, and by-product valorization—that cannot be adequately captured through environmental metrics alone. Criteria selection was informed by LCA outputs, field observations in Bita, and established literature on clean cooking, biomass energy systems, and rural livelihoods, with particular attention to ensuring that each indicator represents a distinct and operational dimension of sustainability relevant to household decision-making.

All criteria were assigned equal weights in the aggregation process. This choice reflects the exploratory purpose of the study and its focus on household “integral security,” where environmental, health, economic, socio-contextual, technical, and systemic dimensions are understood as interdependent in the lived reality of rural households. Although categories contain between three and five criteria (three in most dimensions, four in the socio-contextual category, and five in the environmental category), this distribution reflects the inherently multi-faceted character of environmental performance and related structural constraints. The broader set of environmental indicators captures distinct

biophysical impact pathways rather than assigning disproportionate normative importance to that dimension. The weighting scheme therefore constitutes a transparent normative assumption, consistent with the multi-dimensional nature of energy poverty and the study's objective to avoid privileging a single sustainability objective over household-level priorities.

As a qualitative robustness check, an alternative aggregation structure using equal weights across sustainability categories was examined and did not alter the relative ranking of fuel options, indicating that results are not structurally driven by the distribution of criteria across dimensions. It is nevertheless important to acknowledge that qualitative and semi-quantitative scoring—particularly for socio-contextual and systemic criteria—inevitably involves interpretive judgment. While these assessments were grounded in repeated field observations in Bita and triangulated with relevant literature, they represent structured expert evaluation rather than purely quantitative measurement. The MCDA results should therefore be interpreted as comparative decision-support outputs rather than deterministic rankings. Table 3 documents the data sources and evidentiary basis used to operationalize each MCDA criterion, ensuring transparency and traceability in the scoring process.

Scoring and aggregation

Based on experimental cooking tests reported in Tshimungu et al. (2025), 375 ml of ethanol were required to cook 300 g of rice, compared to approximately 70 g of charcoal and 2 kg of firewood under comparable household conditions. These equivalence values were used to inform MCDA scoring by translating volumetric and mass-based fuels into comparable cooking service units. In addition to fuel quantity, performance indicators such as cooking time, flame stability, smoke exposure, user handling effort, and operating cost were incorporated to capture functional and experiential differences across fuels. Fuel comparison in the MCDA was therefore grounded in delivered cooking service rather than energy-content equivalence, ensuring functional comparability across fuels with differing energy densities and combustion characteristics.

Environmental criteria integrate LCA-derived indicators for ethanol production with literature-based emission factors and comparative benchmarks for firewood and charcoal. Instead of directly transferring absolute LCA impact values into the MCDA, results were converted into relative performance scores to ensure consistency with qualitative and socio-contextual criteria within the decision framework. The LCA was restricted to the cradle-to-gate production stage, whereas the MCDA evaluates full household cooking performance. Combustion efficiency, indoor air pollution exposure, and health-related outcomes were therefore incorporated explicitly at the MCDA stage to maintain analytical coherence and avoid double counting across assessment steps. Health-related indicators were informed by secondary literature on household air pollution and exposure during cooking. Finally, a qualitative robustness check was conducted to assess whether alternative weighting schemes or minor variations in qualitative scoring would substantially alter the relative ranking of fuel options (Table 4).

Treatment of circularity and limitations

Circularity was addressed as a system characteristic embedded in local production and use practices, rather than as a quantified performance metric. In the study area, solid residues from ethanol production—such as fermented pulp and peels—are routinely reused at the local level, notably for soil amendment, construction-related applications, or other household uses. These practices contribute to material recirculation and livelihood support, but remain informal, highly context-specific, and difficult to characterize in terms of avoided products or substitution effects.

Due to this high level of uncertainty, circularity benefits were not credited in the Life Cycle Assessment, which focuses on providing a

Table 3

Data sources and evidentiary basis used for MCDA criteria (revised).

Criterion category	Criterion	Data type	Source type	Evidentiary basis / notes
Environmental	Climate change impact	Quantitative	LCA + literature	LCA results for ethanol (ReCiPe midpoint); comparative emission factors for firewood and charcoal from clean cooking literature (e.g. WHO, IEA).
	Particulate matter formation	Quantitative / qualitative	LCA + literature	Production-stage PM from LCA; combustion-related PM for woodfuels from indoor air pollution studies.
	Human toxicity	Quantitative	LCA + literature	Proxy-based midpoint indicators used comparatively rather than as site-specific health risk estimates.
	Resource depletion	Quantitative / qualitative	LCA + literature	Biomass resource use for ethanol (LCA); forest biomass depletion for firewood and charcoal from forestry literature.
	Water use	Quantitative	LCA + field observations + literature	LCA-derived values primarily reflect observed process water requirements under Bateke Plateau conditions rather than agricultural intensification.
Technical	Combustion cleanliness	Qualitative	Field + literature	Visual smoke, soot deposition, and reported indoor emissions based on field observations and WHO clean cooking benchmarks.
	Energy efficiency	Semi-quantitative	Literature	Relative thermal efficiency of cooking fuels and devices reported in clean cooking and energy access studies.
	Fuel quality consistency	Qualitative	Field observations	Variability in ethanol purity, charcoal quality, and wood moisture under local conditions.
Economic	Household fuel cost	Quantitative	Field observations	Direct monetary expenditure or opportunity cost (time, labor) borne by households.
	Production cost	Semi-quantitative	Field observations	Relative cost structure including labor, water, biomass, and inputs under rural production conditions.
Socio-contextual	Income generation potential	Qualitative	Field observations + literature	Potential for local employment and value addition along the fuel supply chain.
	Feedstock availability	Qualitative	Field observations	Local availability, seasonality, and reliability of raw materials.
	Food security interaction	Qualitative	Literature + field	Degree of competition with food uses (high for maize, low for cassava), based on food security literature and local practices.
	Social acceptance	Qualitative	Field observations	User familiarity, cultural norms, and perceived safety.
	Scalability	Qualitative	Expert judgment + literature	Potential for replication without major infrastructure or institutional change.
Health	Indoor air pollution exposure	Qualitative	Literature + Field observations	Degree of household exposure to harmful emissions during cooking
	Physical effort (drudgery)	Qualitative	Literature + Field observations	Physical workload associated with fuel acquisition, transport, and preparation
	User safety	Qualitative	Field observations	Risk of burns, explosions, or other safety hazards during use
Systemic / circularity	By-product valorization	Qualitative	Field observations	Local reuse of residues (fermentation residues, ash, charcoal fines).
	Pressure on forest resources	Qualitative	Literature + field	Contribution to deforestation and forest degradation through fuel sourcing pathways.
	Dependence on external resources	Qualitative	Field observations + literature	Degree of reliance on non-local inputs (equipment, yeast, water access, supply chains) affecting system autonomy and resilience.

Note: Literature inputs draw on established clean cooking, biomass energy, and forest governance studies (e.g. [Bailis et al., 2015](#); [IEA, 2022](#); [Köhlin et al., 2011](#); [Smith et al., 2014](#); [WHO, 2025](#)). Full references are provided in the reference list. Field observations refer to empirical data collected in Bitá (Bateke Plateau).

Table 4

Scoring scale and normalization rules used in the MCDA.

Score	Interpretation	General meaning across criteria
1	Very poor performance	Very high negative impact, very low efficiency, or very unfavorable condition relative to other options
2	Poor performance	High impact or clear disadvantage
3	Moderate performance	Intermediate or mixed performance
4	Good performance	Low impact or clear advantage
5	Very good performance	Very low impact, high efficiency, or highly favorable condition

conservative and comparable estimate of environmental impacts across ethanol production pathways. Quantifying such benefits within the LCA would have required assumptions about counterfactual uses and long-term environmental outcomes that could not be robustly supported by empirical data. Instead, circularity was explicitly reintroduced in the Multi-Criteria Decision Analysis through qualitative assessment, allowing informal but structurally important practices—such as residue reuse and reduced waste generation—to be considered alongside environmental, technical, economic, and social criteria. This approach enables circularity to be captured as part of the overall sustainability appraisal without overstating its environmental benefits. At the same time, this qualitative treatment also highlights key limitations. In particular, liquid effluents generated during fermentation and distillation are only partially reused and are often discharged untreated, raising concerns

regarding localized environmental impacts and potential toxicity risks. These unresolved issues are therefore considered both a constraint on the circularity of artisanal ethanol systems and a priority area for future improvement.

Study limitations and integration

This study faces limitations inherent to the assessment of informal, small-scale energy systems, particularly with respect to system boundaries and data availability. Rather than attempting further quantification with high uncertainty, these limitations were addressed through an integrative methodological design. The cradle-to-gate scope of the LCA was deliberately selected to enable robust comparison between ethanol production pathways, while impacts related to fuel use and health were incorporated at the MCDA stage.

Similarly, site-specific and informal practices that could not be reliably represented through standardized environmental accounting were integrated qualitatively through MCDA criteria. This approach maintains quantitative rigor where empirical robustness is highest, while ensuring that critical contextual dimensions are not excluded from the overall sustainability appraisal. The resulting framework allows environmental performance to be assessed alongside the socio-economic realities shaping household energy choices in the Democratic Republic of Congo.

Results

Ethanol production performance under rural conditions

This section summarizes the technical performance of the two ethanol production pathways under rural conditions. The production data reported here are derived from field measurements previously documented in Tshimungu et al. (2025) and are briefly recalled to provide contextual grounding for the subsequent environmental and multi-criteria assessments. Detailed production datasets and experimental procedures are therefore not repeated in the present article. The artisanal cassava–maize pathway yielded an average of approximately 25 l of ethanol per batch, corresponding to a conversion efficiency of about 0.40 ml/g of total substrate. Fermentation proceeded spontaneously under ambient conditions, and distillation was conducted over open wood fires without temperature control. As a result, ethanol purity consistently stabilized between 78% and 82% (v/v), reflecting the limited process stability characteristic of artisanal production systems.

The cassava-only pathway was first optimized under laboratory conditions, where controlled fermentation and stable distillation temperatures enabled conversion efficiencies of approximately 0.70 ml/g and ethanol purity levels reaching up to 96% (v/v), establishing a technical performance ceiling for cassava-based ethanol. When this protocol was replicated under field conditions using locally available equipment, performance declined markedly. The replicated batches produced approximately 9 l of ethanol per batch, with conversion efficiency decreasing to around 0.42 ml/g and ethanol purity converging toward artisanal levels (~78% v/v). Across both pathways, ethanol purity converges under rural conditions, indicating that thermal stability during distillation—rather than feedstock composition—is the primary determinant of achievable purity. While the artisanal system delivers higher volumes due to maize supplementation, the cassava-only replication demonstrates that laboratory-derived performance gains cannot be fully transferred without improvements in mixing and heating technologies. These performance characteristics provide the technical baseline for interpreting the environmental and sustainability trade-offs assessed in the subsequent sections.

A simplified energy balance illustrates the importance of distillation efficiency. Assuming an average lower heating value of 16 MJ/kg for firewood, approximately 10 kg of wood correspond to 160 MJ of thermal input. Under artisanal conditions, this quantity yielded 25 l of ethanol, corresponding to roughly 6.4 MJ of wood energy per liter produced, compared to an ethanol calorific value of approximately 21 MJ per liter. This indicates a clearly positive net energy margin. In contrast, under field-replicated conditions, the same quantity of firewood produced only 9 l of ethanol (~17.8 MJ input per liter), substantially narrowing the energy margin. Importantly, the replication retained distillation equipment of identical size despite reduced production scale, increasing fixed thermal losses associated with heating vessel mass and exposed surface area. This demonstrates that net energy performance is strongly determined by equipment scale and thermal design rather than feedstock choice alone.

This study adopts an exploratory design aimed at assessing the environmental feasibility and comparative sustainability performance of cassava-based ethanol under realistic rural conditions. It does not seek to optimize thermodynamic efficiency or maximize production yields. Detailed energy return modeling and system optimization therefore remain important areas for future research.

Environmental performance of ethanol production pathways (LCA)

The life cycle environmental performance of the two cassava-based ethanol production pathways was assessed at both midpoint and endpoint levels using the ReCiPe 2016 impact assessment method. Results are presented to identify dominant impact drivers and to compare relative performance between pathways, while explicitly distinguishing

between proxy-driven effects and site-specific constraints observed in the study area.

Midpoint impact assessment

The midpoint life cycle impact assessment provides a detailed characterization of the environmental pressures associated with the two ethanol production pathways under study: artisanal cassava–maize ethanol and field-replicated cassava-only ethanol. Results are presented per liter of ethanol produced and cover a broad range of impact categories reflecting climate, air pollution, toxicity, resource use, land occupation, and water consumption (Table 5; Fig. 2).

Across both pathways, midpoint impacts are dominated by toxicity- and ecosystem-related categories, particularly terrestrial, freshwater, and marine ecotoxicity, as well as human carcinogenic and non-carcinogenic toxicity. These categories largely shape the overall environmental profile of ethanol production at the production stage. However, given the reliance on proxy agricultural datasets, these toxicity-related results should be interpreted as comparative indicators within the modeled system rather than as direct measures of site-specific environmental or human exposure risk. In contrast, indicators such as stratospheric ozone depletion, ionizing radiation, and particulate matter formation remain comparatively low.

For the artisanal pathway, the most pronounced impacts occur in terrestrial ecotoxicity (4.31 kg 1,4-DCB-eq), human non-carcinogenic toxicity (3.7 kg 1,4-DCB-eq), land use (2.71 m²a crop eq), and global warming (0.763 kg CO₂-eq per liter). These impacts are primarily driven by upstream agricultural processes associated with cassava and maize feedstocks, as represented through proxy datasets, as well as emissions from fermentation residues and wastewater. In the field-replicated cassava-only pathway, the impact profile shifts. While climate change impacts are lower than in the artisanal pathway (0.58 kg CO₂-eq per liter), substantially higher contributions are observed for terrestrial, freshwater, and marine ecotoxicity (10.5, 0.86, and 0.3 kg 1,4-DCB-eq, respectively), along with increased eutrophication indicators. These elevated ecotoxicity values largely reflect pesticide and input assumptions embedded in the Brazilian agricultural proxy datasets rather than measured site-specific conditions in Bitá. They should therefore be interpreted as comparative indicators within the modeled system rather than as direct representations of local environmental toxicity levels. These results reflect the greater relative importance of cassava cultivation and yeast inputs, combined with higher process intensities per liter

Table 5

Environmental results per 1 l if ethanol (artisanal and replication) produced.

Category	Unit	Artisanal	Replication
Global warming	kg CO ₂ -eq	0.763	0.58
Stratospheric ozone depletion	kg CFC11 eq	1.3E-5	6.67E-7
Ionizing radiation	kBq Co-60 eq	0.00879	0.0149
Ozone formation, human health	kg NOx eq	0.00312	0.00293
Fine particulate matter formation	kg PM _{2.5} eq	0.00582	0.00558
Ozone formation, terrestrial ecosystems	kg NOx eq	0.00305	0.00269
Terrestrial acidification	kg SOx eq	0.0146	0.00243
Freshwater eutrophication	kg P eq	0.000329	0.000456
Marine eutrophication	kg N eq	0.00388	0.00158
Terrestrial Ecotoxicity	kg 1,4-DCB-eq	4.31	10.5
Freshwater ecotoxicity	kg 1,4-DCB-eq	0.322	0.86
Marine ecotoxicity	kg 1,4-DCB-eq	0.116	0.3
Human carcinogenic toxicity	kg 1,4-DCB-eq	0.00718	0.00597
Human non-carcinogenic toxicity	kg 1,4-DCB-eq	3.7	1.04
Land use	m ² a crop eq	2.71	2.36
Mineral resource scarcity	kg Cu eq	0.000347	0.00126
Fossil resource scarcity	kg oil eq	0.0933	0.0947
Water consumption	m ³	0.0234	−0.00192

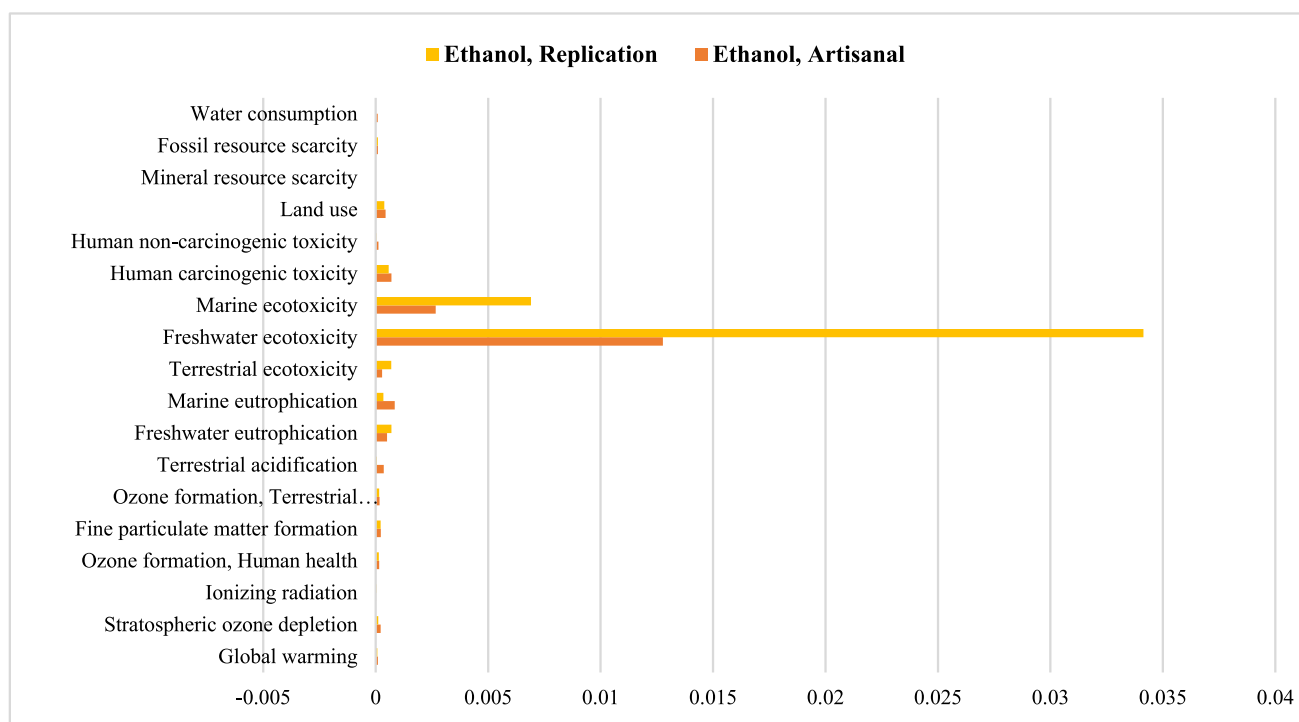


Fig. 2. Weighted results at the midpoint for ethanol (artisanal and replication).

of ethanol produced due to reduced yields under rural replication conditions.

Water consumption exhibits a distinct pattern. Although process-level water requirements are higher in the replication pathway (11 l per liter of ethanol compared to 8 l in the artisanal system), the modeled midpoint water consumption indicator appears negative. This does not indicate a physical reduction in local water use but reflects differences in upstream proxy datasets—particularly feedstock substitution effects within the cradle-to-gate modeling framework. In practical terms, lower yields in the replication pathway increase local water demand per liter of ethanol produced; the negative midpoint value should therefore be interpreted as a modeling artifact rather than a real water saving under rural production conditions. Across both pathways, inputs related to firewood use and biogenic gaseous emissions contribute only marginally to midpoint impacts under standard LCA conventions, as these flows are treated as biogenic within the defined system boundaries.

Overall, neither pathway consistently outperforms the other across all environmental indicators. Instead, each exhibits a distinct profile characterized by trade-offs between climate impacts, toxicity, land occupation, and water-related pressures. These contrasts motivate the subsequent endpoint assessment and integration into the broader multi-criteria appraisal.

Endpoint and aggregated results

The endpoint life cycle impact assessment aggregates midpoint results into three areas of protection—human health, ecosystems, and resource availability—providing a synthesized representation of potential environmental damage associated with each ethanol production pathway. Results are presented using the ReCiPe Endpoint (H) method after normalization and weighting (Fig. 3).

Across both pathways, damage to human health dominates the aggregated environmental profile. The artisanal pathway exhibits higher human health damage (87.3 mPt) than the replication pathway (71.5 mPt), largely driven by toxicity-related midpoint categories and particulate emissions. Ecosystem damage follows a similar pattern, with higher impacts for the artisanal pathway (8.22 mPt) compared to the replication pathway (6.62 mPt). Resource scarcity contributes only

marginally to total damage for both pathways. While aggregation reduces the dimensional complexity observed at the midpoint level, it clarifies the dominant trade-offs between production systems.

The replication pathway achieves lower aggregated damage to human health and ecosystems, despite higher midpoint contributions in certain ecotoxicity categories and greater process water intensity per liter of ethanol produced. Conversely, the artisanal pathway shows comparatively lower ecotoxicity at the midpoint level but higher overall endpoint damage due to its combined contributions across multiple human health and land-use-related categories.

These results confirm that neither pathway is uniformly superior across all environmental dimensions. Instead, each represents a distinct configuration of trade-offs. For the purpose of the subsequent multi-criteria decision analysis—where ethanol is compared against alternative cooking fuels—the replication pathway was retained as the representative ethanol configuration. This choice reflects its lower aggregated endpoint damage and its relevance as a feedstock-substitution strategy under rural conditions. It does not imply overall environmental optimality, but provides a transparent and consistent basis for integrating environmental performance into a broader sustainability appraisal that also considers technical, economic, social, and systemic constraints (Fig. 3).

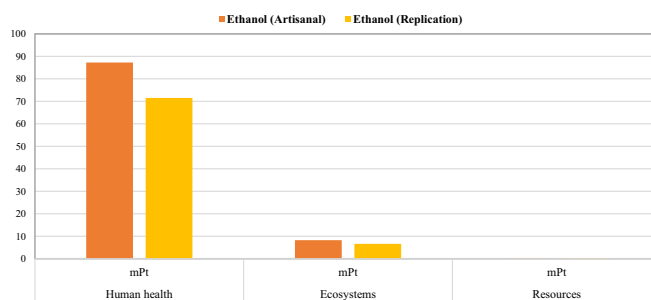


Fig. 3. Impacts of ethanol production at the areas of protection (human health, ecosystems, and resources).

Comparative sustainability appraisal of cooking fuels (MCDA)

This section presents the results of the multi-criteria decision analysis (MCDA) comparing three household cooking fuel options under current rural conditions in Bita: firewood, charcoal, and ethanol. In line with the environmental assessment, ethanol refers to the cassava-only replicated production pathway, selected based on its lower aggregated endpoint damage and its relevance for feedstock substitution.

MCDA scoring matrix and aggregated results

Table 6 reports the MCDA scoring matrix, aggregated scores, and composite impact scores for the three fuel options across environmental, technical, economic, health, socio-contextual, systemic, and physical-effort dimensions. Scores reflect relative performance based on the criteria and scoring rules defined in the methodology and are aggregated using a simple additive weighting approach with equal weights assigned to all criteria.

The aggregated MCDA scores indicate that ethanol achieves the highest overall score (67), followed by charcoal (57) and firewood (55). When expressed as composite impact scores on a 1–5 scale, ethanol reaches an average score of 3.19, compared to 2.71 for charcoal and 2.62 for firewood. These results highlight a differentiated sustainability performance among fuel options rather than the dominance of a single option across all criteria.

Category-level performance profiles

At the category level, ethanol demonstrates consistently strong performance in health-related criteria, including indoor air pollution exposure, user safety, and physical effort. These results reflect its cleaner combustion characteristics and reduced labor burden at the household level compared to traditional woodfuels. Ethanol also performs well on systemic criteria, particularly pressure on forest resources and by-product valorization, indicating a lower structural dependence on forest biomass and greater potential for local material reuse. Fig. 4

Table 6
MCDA scoring matrix and aggregated scores (1 = very poor; 5 = very good).

Criterion category	Criterion	Firewood	Charcoal	Ethanol
Environmental	Climate change impact	2	1	3
	Particulate matter formation	1	2	4
	Human toxicity	2	2	3
	Resource depletion	2	1	4
	Water use	5	4	2
Technical	Combustion cleanliness	1	2	4
	Energy efficiency	2	3	3
	Fuel quality consistency	2	3	3
	Household fuel cost	5	3	2
Economic	Production cost	4	3	2
	Income generation potential	2	3	4
	Feedstock availability	5	4	3
Socio-contextual	Food security interaction	5	5	3
	Social acceptance	2	4	4
	Scalability	5	4	2
Health	Air pollution exposure	1	2	4
	Physical effort burden (drudgery)	1	3	4
Systemic / circularity	User safety	1	2	3
	By-product valorization	2	2	4
	Pressure on forest resources	1	1	4
	Dependence on external resources	4	3	2
	Aggregated MCDA score (sum)	55	57	67
Composite impact score (average, 1–5)		2.62	2.71	3.19

illustrates the category-level performance profiles of the three cooking fuel options based on average MCDA scores (1–5) across environmental, technical, economic, socio-contextual, health, and systemic dimensions. The radar plot highlights distinct performance profiles across fuel options. Ethanol exhibits a relatively balanced profile, with strong performance in health, technical, and systemic dimensions, and moderate performance in environmental and socio-contextual categories. Firewood and charcoal display more uneven profiles, characterized by strong socio-contextual and economic performance but weak outcomes in health and environmental dimensions. Charcoal performs better than firewood in technical and health-related categories, while firewood retains advantages in feedstock availability and economic accessibility.

The visualization emphasizes that differences between options are not driven by a single dominant category, but by contrasting configurations of strengths and weaknesses across dimensions.

Category scores represent averages of equally weighted criteria and are intended to illustrate relative performance patterns rather than overall rankings.

Charcoal occupies an intermediate position across most categories. It performs better than firewood in technical criteria such as combustion cleanliness and fuel quality consistency, as well as in physical effort and social acceptance. However, these advantages are offset by poor performance on environmental and systemic criteria related to forest resource depletion and upstream inefficiencies.

Firewood exhibits a contrasting profile. It scores highly on feedstock availability, household fuel cost, and low dependence on external resources, reflecting its accessibility and simplicity under rural conditions. At the same time, it is strongly penalized in health, technical, and environmental categories due to high particulate emissions, elevated physical effort, and significant pressure on forest resources.

Trade-offs and decision-relevant contrasts

The MCDA results reveal that the relative ranking of fuel options is driven by trade-offs between short-term accessibility and long-term sustainability outcomes. Firewood and charcoal remain competitive primarily because of their affordability, availability, and integration into existing practices, rather than their environmental or health performance. Ethanol, by contrast, achieves a higher overall score by improving health outcomes, reducing labor burdens, and alleviating forest pressure, despite clear constraints related to water use, production cost, and scalability. These findings indicate that ethanol does not represent a universally superior solution, but rather a conditional improvement pathway whose advantages depend on addressing specific technical and systemic constraints. The MCDA thus highlights the importance of evaluating cooking fuel transitions through a multi-dimensional lens that captures environmental impacts alongside health, labor, economic, and contextual factors relevant to household decision-making.

Discussion

Ethanol within biomass lock-in and sustainable development trade-offs

The persistence of traditional biomass fuels in rural energy systems is widely interpreted as a socio-technical lock-in shaped by poverty, institutional inertia, and daily survival constraints (Creutzig et al., 2015; Geels, 2004; Köhlin et al., 2011). In the Congolese context, this biomass lock-in reflects structural constraint rather than voluntary preference (Tshimungu et al., 2025). The results of this study substantiate this perspective, demonstrating that firewood and charcoal perform strongly in socio-contextual and economic dimensions not because they are sustainable, but because they are accessible and deeply embedded in local livelihood systems. As noted by Puzzolo et al. (2016) and Mortimer et al. (2017), technological superiority alone rarely disrupts such structural lock-ins.

Cassava-based ethanol does not eliminate biomass dependence but



Fig. 4. Category-level MCDA performance profiles of firewood, charcoal, and ethanol based on average scores (1–5) across sustainability dimensions.

reconfigures it. By shifting from direct wood combustion to locally produced liquid biofuel, it reduces forest pressure while remaining within the broader biomass economy. From a transition perspective, ethanol can thus be understood as a niche innovation operating within the dominant biomass regime (Geels, 2004). Its contribution lies not in replacing biomass entirely, but in transforming it into a cleaner and more controllable energy carrier.

However, this repositioning raises broader sustainable development considerations. The global biofuel literature has long debated the food–fuel trade-off associated with first-generation biofuels (HLPE, 2013; Searchinger et al., 2008). In Bita, cassava functions simultaneously as a food crop and income source, while the artisanal pathway also involves maize, water, and firewood inputs. Although the cassava-only replication pathway reduces maize dependence and partially limits direct food substitution effects, ethanol production remains embedded within agrarian systems characterized by constrained land productivity and seasonal water scarcity.

Scalability is therefore structurally constrained. Technical limitations in distillation efficiency, seasonal water fluctuations on the Bateke Plateau, and the absence of cooperative structures or accessible micro-financing mechanisms limit expansion beyond small-scale production. As Sovacool and Geels (2016) emphasize, niche innovations in low-income contexts rarely scale without targeted institutional support. Cassava-based ethanol should thus be interpreted as a context-bound improvement pathway rather than a transformative substitute, capable of alleviating key harms of biomass lock-in while remaining subject to structural constraints.

Interpreting environmental trade-offs and circularity in context

The LCA results highlight the limits of single-indicator sustainability assessments (D'Amato et al., 2017; Finnveden et al., 2009). Although the cassava-only pathway shows lower aggregated endpoint damage, it entails higher midpoint impacts in certain ecotoxicity categories and

greater process water intensity. These contrasts illustrate that sustainability outcomes depend on how environmental trade-offs are interpreted rather than on isolated indicators.

Toxicity-related categories dominate midpoint results, largely driven by agricultural proxy datasets rather than measured site-specific conditions. These indicators should therefore be understood as comparative signals within the modeled system rather than as direct representations of local environmental or human exposure risk. Similarly, the negative water consumption value observed in the replication pathway reflects modeling effects linked to upstream substitutions rather than a physical reduction in local water use. Field observations indicate that lower yields increase local water demand per liter of ethanol produced. As Boulay et al. (2018) argue, midpoint water indicators often require contextual interpretation to avoid misleading conclusions.

Circularity further illustrates the contextual nature of environmental assessment. Residue reuse in ethanol production demonstrates meaningful material recirculation practices embedded in informal systems. However, these practices remain highly context-specific and difficult to quantify within conventional LCA boundaries (HLPE, 2020; Kirchherr et al., 2017). Wastewater management and effluent discharge reveal the ambivalent character of informal circularity: while certain by-products are valorized locally, unresolved liquid effluent management raises concerns regarding localized environmental risks. Recognizing both adaptive capacity and structural limits avoids romanticizing informality and underscores that environmental improvements remain conditional on improved process management.

Lived experience, fuel stacking, and policy-relevant implications

A central insight of this study is that sustainability performance cannot be reduced to environmental efficiency alone. Consistent with WHO (2025) and Bonjour et al. (2013), firewood performs poorly in health-related criteria due to its association with indoor air pollution and respiratory risk. The MCDA further demonstrates that physical

effort—often neglected in clean cooking assessments—plays a decisive role in shaping household perceptions of fuel desirability.

In Bitá, fuel choices are driven less by calorific value than by the desire to reduce daily hardship (Tshimungu et al., 2025). Ethanol's strong performance in health, safety, and labor-related criteria aligns with these lived priorities, reinforcing arguments that energy transitions gain traction when they deliver tangible improvements in everyday experience (Sovacool, 2016). However, consistent with clean cooking literature, full fuel substitution is unlikely. Households are expected to engage in fuel stacking behavior, combining ethanol with wood-based fuels depending on availability, cost, and specific cooking tasks (Masera et al., 2000). Ethanol should therefore be understood as a partial displacement fuel rather than a complete replacement.

These findings carry concrete policy implications. Clean cooking strategies in low-income rural contexts should move beyond binary distinctions between “traditional” and “modern” fuels. Transitional solutions that reduce health burdens and forest pressure without requiring immediate structural transformation may offer more realistic pathways under conditions of energy poverty (Köhlin et al., 2011).

Targeted technical support to improve distillation efficiency—particularly through improved insulation, vessel sizing, and thermal control—could reduce water intensity and enhance net energy performance without altering feedstock systems. Decentralized ethanol initiatives should be integrated with local water governance and agricultural productivity strategies to mitigate resource competition. While the findings are context-specific, similar biomass-dependent rural systems across Sub-Saharan Africa face comparable structural constraints, suggesting cautious but meaningful potential for contextual transferability.

Conclusion

This study demonstrates that improving household cooking systems in rural contexts requires moving beyond binary distinctions between traditional and modern fuels. By combining Life Cycle Assessment and Multi-Criteria Decision Analysis, the analysis shows that cassava-based ethanol does not constitute a universal solution, but rather a context-dependent improvement pathway within biomass-dependent energy systems.

Compared with firewood and charcoal, ethanol offers clear advantages in terms of health, physical effort, and pressure on forest resources, directly addressing some of the most severe burdens associated with traditional biomass use. However, these benefits are accompanied by new constraints related to water use, production efficiency, cost, and scalability, which limit its transformative potential under current conditions.

Rather than framing ethanol as a full replacement fuel, the findings suggest that it should be understood as a locally embedded niche innovation capable of alleviating key harms of biomass lock-in when supported by appropriate technical and institutional arrangements. More broadly, the study highlights the importance of decision-oriented, multi-dimensional assessment frameworks for evaluating energy transitions in low-income settings, where sustainability outcomes emerge from trade-offs shaped by lived realities as much as by environmental performance metrics.

A practical policy priority emerging from this analysis is the introduction of improved, appropriately scaled distillation units designed to reduce thermal losses under rural conditions. Given that net energy performance is strongly influenced by equipment size and heat management rather than feedstock composition alone, technical support focusing on insulation, vessel sizing, and thermal control could significantly enhance system efficiency without requiring structural changes to feedstock systems. Such incremental improvements may yield greater sustainability gains than attempting rapid fuel substitution without addressing production efficiency.

CRedit authorship contribution statement

Theresia Besala Tshimungu: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mohammed Engsha Isah:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation. **Keiichi Ishii:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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